Jiten Rai

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Education

Texas College of Management and IT — Mitrapark, Kathmandu
Bachelor of Information Technology (Hons) — *2022 – 2026*

Xavier International College — Kalopul, Kathmandu
+2 Levels Science (Physics) — *2015 – 2017*

Work Experience

Computer Vision Intern — May 2025 - Aug 2025
Danson Solutions — *Gairidhara, Kathmandu*

- Worked on multiple mini projects on Computer Vision, YOLO, and CNN
- Performed on prolific standards to lead a team on separate interactive game development project
- Explored the concepts of Gen AI and Fast API endpoint generation for delivering a backend module

AI Intern — Nov 2025 - Jan 2026
Frost Digital Ventures — *Lampati, Kalanki*

- Accomplished multiple weekly projects on EDA, ML algorithms, CNN, LLM and RAG.
- Delivered in detailed submission of tasks with calibration ablation on NN models, finetuning with LoRA-PEFT, evaluation reports on MRR and nDCG, graph structure manipulation using Networkx and Neo4j.
- Logging practice implied works with MLflow and WandB.
- Complete copilot application development with VS code extension delivery.

Projects

Projector Camera Interactive Balloon Pop Game | LINK | *Python, Pygame, OpenCV, Ultralytics YOLO, ONNX*

- An interactive balloon-popping game that uses computer vision (YOLO object detection via ONNX), OpenCV, and Pygame to detect and respond to real-world interactions projected onto a screen, with threading for real-time camera capture and smooth gameplay
- The system supports camera calibration, perspective correction, and multi-monitor/projector setups, and features animated visuals, sound effects, and score tracking, with assets managed in a structured directory
- The technical stack includes Python, Pygame for graphics and event handling, OpenCV for camera and image processing, Ultralytics YOLO for object detection, and threading for efficient parallel camera frame acquisition

QuickDraw Game - AI-Powered Drawing Recognition | LINK | *Python, FastAPI, TensorFlow, Vanilla JS, Keras*

- Full-stack AI-powered drawing game: users draw objects on a smooth, responsive Vanilla JS canvas with real-time feedback, recognized by a CNN model integrated via FastAPI and Keras/TensorFlow for multi-class classification and confidence calibration.
- Comprehensive API endpoints for drawing recognition, random object selection, and model info; model training and calibration handled in Jupyter notebooks, with detailed results, confidence scores, error handling, mobile support, and clear project structure for easy deployment.

Developer Copilot: AI-Powered RAG and PR Summarization | LINK | *Python, ChromaDB, Groq, RAG, VS Code Extension*

- An intelligent developer productivity suite featuring code search and pull request summarization, leveraging advanced natural language processing, vector search, and Retrieval-Augmented Generation (RAG) for comprehensive, context-aware codebase navigation and review.
- Implements PR summarization by executing Git commands to extract atomic code changes, enabling precise, commit-level analysis and generating concise, context-rich summaries for each pull request.
- The technical stack includes Python for backend logic, ChromaDB for vector storage, Groq API (Llama-3.3-70B-Versatile) for language understanding, RAG for improved search, and TypeScript for VS Code extension development, with structured assets and documentation for easy deployment and extension.

MCQ Paper Grader | LINK | *Python, OpenCV, Streamlit, Pandas*

- Developed a web-based application to automate grading of multiple-choice answer sheets using image processing
- Utilized OpenCV for bubble detection, Streamlit for an interactive user interface, and Pandas for result analysis and visualization

Certifications

AI Internship Experience Letter | LINK | Frost Digital Ventures — Jan 2026
Computer Vision Internship Letter | LINK | Danson Solutions — Aug 2025
Deep Learning Specialization | LINK | Coursera — Feb 2025
Sequence Models | LINK | DeepLearning.AI — March 19, 2025
Convolutional Neural Network | LINK | DeepLearning.AI — March 10, 2025
Structuring Machine Learning Projects | LINK | DeepLearning.AI — March 4, 2025
Improving Deep Neural Networks: Hyperparameter Tuning, Regularization | LINK | DeepLearning.AI — February 27, 2025
Neural Networks and Deep Learning | LINK | DeepLearning.AI — February 13, 2025

Skills Summary

Programming: Python, Data Structures, Algorithms

Libraries/Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow

Domains: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Science

Tools: Git, Jupyter Notebook, Virtual Environments, VS Code

Soft Skills: Problem-Solving, Data Analysis, Project Management, Team Collaboration

Additional Information

Languages: English (Fluent), Nepali (Native)

Awards/Activities:

- Winner, Texas Dev Rumble 2025 - One Day Code Competition, Collaborated with a team developed an end-to-end web application on virtual study rooms with session based context level understanding chatbot feature
- Winner, Texas Hackathon 2025 - Volume I, Collaborated with my team to develop a virtual assistant for visually impaired students
- Robotics Expo - Sep 2025, Presented a group project on a Remote-Controlled IR Ray Sensor Light built using Arduino and IoT concepts

References

References available upon request.